

Week 5 Reading

Valuation, NPV, Benefit-Cost Analysis

Draft Material — This content is under development and subject to change.

1 Assignment

Read K. Arrow *et al.* [1]. Focus especially on Sections 1-3 (the two rationales for discounting and the argument for declining discount rates).

If you want additional context on how the US government practices BCA, skim U.S. Office of Management and Budget [2] (Sections I-III are most relevant).

2 Overview

Monday's lecture introduced benefit-cost analysis as a framework for comparing decisions under uncertainty. A key component is the discount rate: how much less we value future costs and benefits compared to present ones. This choice is especially consequential for climate adaptation decisions with long time horizons.

Arrow et al. (2013) review two rationales for discounting (consumption-based and investment-based) and argue that uncertainty about future discount rates leads to an effectively declining discount rate over long horizons. The White House Circular A-94 (2023) provides the current federal guidance for how BCA should be practiced in the US government.

3 Discussion Questions

1. Arrow et al. describe two rationales for discounting (consumption-based and investment-based). Which do you find more convincing for climate adaptation decisions, and why?
2. If discount rates are uncertain, Arrow et al. argue this is "equivalent to a declining discount rate." What are the practical implications of this for a 50-year infrastructure project?
3. The White House Circular A-94 represents how the US government *actually* practices BCA. Identify one place where the guidance addresses (or fails to address) a critique we discussed in lecture.
4. Consider your final project topic. What costs and benefits would you need to quantify? Which would be hardest to monetize?

4 References

Bibliography

- [1] K. Arrow *et al.*, "Determining Benefits and Costs for Future Generations," *Science*, vol. 341, no. 6144, pp. 349–350, Jul. 2013, doi: 10.1126/science.1235665.

- [2] U.S. Office of Management and Budget, "Circular No. A-94: Guidelines and Discount Rates for Benefit-Cost Analysis of Federal Programs," Nov. 2023. [Online]. Available: <https://www.whitehouse.gov/wp-content/uploads/2023/11/CircularA-94.pdf>